An Arbitrary-Norm Vector Contraction Inequality
and Its Scope for Predict-Then-Optimize Generalization Bounds

Clean verification note

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Abstract

Status. This note gives a complete solution of the standalone arbitrary-norm vector-contraction subproblem for the standard coordinate Rademacher multivariate complexity. The constant obtained below is finite for every norm on finite-dimensional $\mathbb{R}^d$, and it is optimal for that comparison. Relative to the predict-then-optimize paper of El Balghiti, Elmachtoub, Grigas, and Tewari, the result should be described as partial progress rather than as a solution of the whole paper: it supplies a rigorous replacement for the Euclidean vector-contraction step in the margin-based argument, but it does not resolve the other parts of the predict-then-optimize theory, such as the construction and computation of the margin SPO loss or the verification of the strength property.

Let $(\mathbb{R}^d, \|\cdot\|)$ be any finite-dimensional normed space. For a symmetric random dual vector Y, put

$$\rho(Y; \|\cdot\|) := \inf_{v \neq 0} \frac{\mathbb{E}\,|\langle Y, v\rangle|}{\|v\|}.$$

If $\rho(Y; \|\cdot\|) > 0$, then every scalar L-Lipschitz post-composition of a vector-valued class has empirical Rademacher complexity at most L/ρ times the corresponding Y-linear multivariate complexity. Taking Y to be a vector of independent coordinate Rademacher signs gives

$$\widehat{\mathfrak{R}}_n(h \circ \mathcal{H}) \leq \frac{L}{\kappa_{\|\cdot\|}} \widehat{\mathfrak{R}}_n(\mathcal{H}), \qquad \kappa_{\|\cdot\|} := \inf_{v \neq 0} \frac{\mathbb{E}_\tau\,|\langle \tau, v\rangle|}{\|v\|} > 0.$$

The constant $\kappa_{\|\cdot\|}^{-1}$ is the best possible constant in this inequality, uniformly over all samples and all vector-valued classes. For ℓ_q^d it implies the explicit bound

$$\widehat{\mathfrak{R}}_n(h \circ \mathcal{H}) \leq \sqrt{2}\, L\, d^{(1/q - 1/2)_+} \widehat{\mathfrak{R}}_n(\mathcal{H}), \qquad 1 \leq q \leq \infty,$$

with the sharper value $\kappa_{\ell_\infty^d}^{-1} = 1$ for the full multivariate Rademacher complexity. For $1 \leq q < 2$, a dimension factor of order $d^{1/q - 1/2}$ is unavoidable if one insists on comparing to the standard coordinate-Rademacher complexity. If one instead permits norm-adapted linear complexities, the same theorem gives dimension-free q-stable comparisons for $1 < q < 2$ and an exact coordinate-sampling comparison for $q = 1$.

Contents

1 What is solved, and what is only partial progress

The technical issue isolated here is the following vector-contraction problem. Let $x_1, \dots, x_n \in \mathcal{X}$ be fixed and let $\mathcal{H}$ be a class of maps $f : \mathcal{X} \to \mathbb{R}^d$. The standard empirical multivariate Rademacher complexity is

$$\widehat{\mathfrak{R}}_n(\mathcal{H}; x_{1:n}) := \mathop{\mathbb{E}}_{\sigma} \left[\sup_{f \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^{n} \langle \sigma_i, f(x_i) \rangle \right], \tag{1}$$

where the vectors $\sigma_1, \dots, \sigma_n \in \{\pm 1\}^d$ have independent Rademacher coordinates. Given scalar maps $h_i : \mathbb{R}^d \to \mathbb{R}$, one wants to compare

$$\mathop{\mathbb{E}}_{\varepsilon} \left[\sup_{f \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^{n} \varepsilon_i h_i(f(x_i)) \right] \tag{2}$$

to (1) when the maps h_i are Lipschitz with respect to a norm other than the Euclidean norm.

Maurer's theorem gives such a comparison, with constant $\sqrt{2}L$, when the h_i are L-Lipschitz in the Euclidean norm [4]. El Balghiti, Elmachtoub, Grigas, and Tewari use this kind of step in their margin-based predict-then-optimize generalization bounds, and explicitly note that their margin bounds are stated in the Euclidean setting because the available vector-contraction lemma they use is Euclidean [2]. The present note proves the arbitrary-norm analogue of this step.

It is important not to overstate the conclusion. The result below is a complete answer to the contraction question above, including the sharp constant for the standard coordinate-Rademacher process. It is not a full solution of the predict-then-optimize paper. The paper also studies the SPO loss, margin surrogates, geometric strength assumptions, and computational issues; none of those are solved here. What is obtained is a plug-in replacement for one analytic ingredient: the passage from a Lipschitz scalar loss class to the multivariate complexity of the vector-valued prediction class.

All results are first stated for fixed samples. Measurability issues are not central to the argument. One may either assume finite classes throughout, or interpret suprema by outer expectation and obtain the stated inequalities by the usual finite-subclass approximation.

2 A first-moment replacement lemma

Let $E = (\mathbb{R}^d, \|\cdot\|)$ be a finite-dimensional normed space. We identify the dual vector space with $\mathbb{R}^d$ through the standard pairing $\langle y, v \rangle = \sum_{j=1}^{d} y_j v_j$.

Definition 2.1 (First-moment domination constant)**.** *Let Y be a symmetric random dual vector, meaning Y and $-Y$ have the same distribution. Define*

$$\rho(Y; \|\cdot\|) := \inf_{v \neq 0} \frac{\mathbb{E}\,|\langle Y, v\rangle|}{\|v\|}. \tag{3}$$

Equivalently, $\rho(Y; \|\cdot\|)$ is the largest $\rho \geq 0$ such that

$$\rho\,\|v\| \leq \mathbb{E}\,|\langle Y, v\rangle| \qquad \text{for all } v \in E. \tag{4}$$

Lemma 2.2 (One-coordinate replacement)**.** *Let S be a set, let $a : S \to \mathbb{R}$, $\psi : S \to \mathbb{R}$, and $\varphi : S \to E$. Suppose that*

$$|\psi(s) - \psi(t)| \leq L\,\|\varphi(s) - \varphi(t)\| \qquad \text{for all } s, t \in S. \tag{5}$$

Let Y be symmetric and let $\rho := \rho(Y; \|\cdot\|) > 0$. If ε is an independent scalar Rademacher sign, then

$$\mathop{\mathbb{E}}_{\varepsilon} \sup_{s \in S} \{\varepsilon\psi(s) + a(s)\} \leq \mathop{\mathbb{E}}_{Y} \sup_{s \in S} \left\{ \frac{L}{\rho} \langle Y, \varphi(s)\rangle + a(s) \right\}. \tag{6}$$

Proof. We give the proof for finite S; the general case follows by outer expectations or finite-subclass approximation. First,

$$\begin{aligned}
\mathop{\mathbb{E}}_{\varepsilon} \sup_{s} \{\varepsilon\psi(s) + a(s)\} &= \frac{1}{2} \sup_{s} \{\psi(s) + a(s)\} + \frac{1}{2} \sup_{t} \{-\psi(t) + a(t)\} \\
&= \frac{1}{2} \sup_{s,t} \{\psi(s) - \psi(t) + a(s) + a(t)\} \\
&\leq \frac{1}{2} \sup_{s,t} \{L\,\|\varphi(s) - \varphi(t)\| + a(s) + a(t)\}.
\end{aligned}$$

By (4), for every $u \in E$,

$$L\,\|u\| \leq \frac{L}{\rho} \mathop{\mathbb{E}}_{Y} |\langle Y, u\rangle|.$$

Thus, using $\sup \mathbb{E}\, B_{s,t}(Y) \leq \mathbb{E} \sup B_{s,t}(Y)$,

$$\mathop{\mathbb{E}}_{\varepsilon} \sup_{s} \{\varepsilon\psi(s) + a(s)\} \leq \frac{1}{2} \mathop{\mathbb{E}}_{Y} \sup_{s,t} \left\{ \frac{L}{\rho} |\langle Y, \varphi(s) - \varphi(t)\rangle| + a(s) + a(t) \right\}.$$

For fixed Y, the supremum over ordered pairs is unchanged if the absolute value is removed. Indeed, if $\langle Y, \varphi(s) - \varphi(t)\rangle < 0$, then swapping s and t changes the sign and leaves $a(s) + a(t)$ unchanged. Hence the last display equals

$$\begin{aligned}
\frac{1}{2} \mathop{\mathbb{E}}_{Y} \sup_{s,t} \left\{ \frac{L}{\rho} \langle Y, \varphi(s) - \varphi(t)\rangle + a(s) + a(t) \right\} = \frac{1}{2} \mathop{\mathbb{E}}_{Y} \Bigg[&\sup_{s} \left\{ \frac{L}{\rho} \langle Y, \varphi(s)\rangle + a(s) \right\} \\
&+ \sup_{t} \left\{ -\frac{L}{\rho} \langle Y, \varphi(t)\rangle + a(t) \right\} \Bigg].
\end{aligned}$$

The two expectations on the right are equal because $Y \stackrel{d}{=} -Y$. This proves (6). ∎

3 The arbitrary-norm vector-contraction theorem

Theorem 3.1 (Distributional vector contraction). *Let S be a set and let $E = (\mathbb{R}^d, \|\cdot\|)$. For $i = 1, \ldots, n$, let $\psi_i : S \to \mathbb{R}$ and $\varphi_i : S \to E$ satisfy*

$$|\psi_i(s) - \psi_i(t)| \le L_i \|\varphi_i(s) - \varphi_i(t)\| \qquad \text{for all } s, t \in S. \tag{7}$$

Let Y be a symmetric random dual vector with $\rho := \rho(Y; \|\cdot\|) > 0$, and let $Y_1, \ldots, Y_n$ be independent copies of Y. Then

$$\mathop{\mathbb{E}}_{\varepsilon} \sup_{s \in S} \sum_{i=1}^{n} \varepsilon_i \psi_i(s) \le \mathop{\mathbb{E}}_{Y} \sup_{s \in S} \sum_{i=1}^{n} \frac{L_i}{\rho} \langle Y_i, \varphi_i(s) \rangle . \tag{8}$$

In particular, if $L_i \le L$ for all i, then

$$\mathop{\mathbb{E}}_{\varepsilon} \sup_{s \in S} \sum_{i=1}^{n} \varepsilon_i \psi_i(s) \le \frac{L}{\rho} \mathop{\mathbb{E}}_{Y} \sup_{s \in S} \sum_{i=1}^{n} \langle Y_i, \varphi_i(s) \rangle . \tag{9}$$

Proof. For $m = 0, 1, \ldots, n$, define

$$Q_m := \mathbb{E} \sup_{s \in S} \left\{ \sum_{i=1}^{m} \frac{L_i}{\rho} \langle Y_i, \varphi_i(s) \rangle + \sum_{i=m+1}^{n} \varepsilon_i \psi_i(s) \right\},$$

where the expectation is over all variables appearing in the display. We prove $Q_{m-1} \le Q_m$ for each m. Condition on all random variables except ε_m and Y_m, and write

$$a(s) = \sum_{i=1}^{m-1} \frac{L_i}{\rho} \langle Y_i, \varphi_i(s) \rangle + \sum_{i=m+1}^{n} \varepsilon_i \psi_i(s).$$

With this conditionally fixed offset a, Lemma 2.2 applied to ψ_m and φ_m gives the conditional inequality that replaces $\varepsilon_m \psi_m$ by $(L_m/\rho) \langle Y_m, \varphi_m \rangle$. Taking expectations yields $Q_{m-1} \le Q_m$. Iterating from $m = 1$ to n gives $Q_0 \le Q_n$, which is (8). If $L_i \le L$, then (7) also holds with the common constant L in every coordinate, and (9) follows from the already proved case. □

For a fixed sample $x_{1:n}$ and a symmetric dual random vector Y, define the Y-linear empirical complexity

$$\widehat{\mathfrak{R}}_n^Y(\mathcal{H}; x_{1:n}) := \mathop{\mathbb{E}}_{Y} \left[\sup_{f \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^{n} \langle Y_i, f(x_i) \rangle \right]. \tag{10}$$

Theorem 3.1 immediately gives the following form.

Corollary 3.2 (Post-composition form). *Let $h_i : (\mathbb{R}^d, \|\cdot\|) \to \mathbb{R}$ be L_i-Lipschitz, let Y be symmetric, and suppose $\rho(Y; \|\cdot\|) > 0$. Then*

$$\mathop{\mathbb{E}}_{\varepsilon} \left[\sup_{f \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^{n} \varepsilon_i h_i(f(x_i)) \right] \le \frac{1}{n} \mathop{\mathbb{E}}_{Y} \left[\sup_{f \in \mathcal{H}} \sum_{i=1}^{n} \frac{L_i}{\rho(Y; \|\cdot\|)} \langle Y_i, f(x_i) \rangle \right]. \tag{11}$$

If $L_i \le L$ for all i, then

$$\mathop{\mathbb{E}}_{\varepsilon} \left[\sup_{f \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^{n} \varepsilon_i h_i(f(x_i)) \right] \le \frac{L}{\rho(Y; \|\cdot\|)} \widehat{\mathfrak{R}}_n^Y(\mathcal{H}; x_{1:n}). \tag{12}$$

4 The standard Rademacher comparison and its optimal constant

Take $Y = \tau$, where $\tau = (\tau_1, \dots, \tau_d)$ has independent coordinate Rademacher signs. Define

$$\kappa_{\|\cdot\|} := \rho(\tau; \|\cdot\|) = \inf_{v \neq 0} \frac{\mathbb{E}_\tau |\langle \tau, v\rangle|}{\|v\|} = \min_{\|v\|=1} 2^{-d} \sum_{\tau \in \{\pm 1\}^d} |\langle \tau, v\rangle| . \tag{13}$$

This number is positive. The unit sphere of $\|\cdot\|$ is compact, the objective in (13) is continuous, and it is positive on nonzero v: if $\langle \tau, v\rangle = 0$ for every sign vector τ, then

$$0 = 2^{-d} \sum_{\tau \in \{\pm 1\}^d} \tau_j \langle \tau, v\rangle = v_j \qquad (j = 1, \dots, d),$$

so $v = 0$.

Theorem 4.1 (Arbitrary-norm Rademacher contraction)**.** *Let $h_i : (\mathbb{R}^d, \|\cdot\|) \to \mathbb{R}$ be L_i-Lipschitz. Then, for every vector-valued class $\mathcal{H}$ and every fixed sample $x_{1:n}$,*

$$\mathop{\mathbb{E}}_{\varepsilon} \left[\sup_{f \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^{n} \varepsilon_i h_i(f(x_i)) \right] \leq \frac{1}{n} \mathop{\mathbb{E}}_{\sigma} \left[\sup_{f \in \mathcal{H}} \sum_{i=1}^{n} \frac{L_i}{\kappa_{\|\cdot\|}} \langle \sigma_i, f(x_i)\rangle \right]. \tag{14}$$

In particular, if h is L-Lipschitz and $h_i = h$ for all i, then

$$\widehat{\mathfrak{R}}_n(h \circ \mathcal{H}; x_{1:n}) \leq \frac{L}{\kappa_{\|\cdot\|}} \widehat{\mathfrak{R}}_n(\mathcal{H}; x_{1:n}). \tag{15}$$

Proof. This is Corollary 3.2 with $Y = \tau$. In the common Lipschitz-constant case, the right-hand side is exactly the standard multivariate Rademacher complexity (1) multiplied by $L/\kappa_{\|\cdot\|}$. □

The constant in (15) is not an artifact of the proof.

Proposition 4.2 (Sharpness for the standard comparison)**.** *For a fixed norm $\|\cdot\|$ on $\mathbb{R}^d$, the smallest constant C such that*

$$\widehat{\mathfrak{R}}_n(h \circ \mathcal{H}; x_{1:n}) \leq CL\widehat{\mathfrak{R}}_n(\mathcal{H}; x_{1:n}) \tag{16}$$

holds for every n, every sample, every vector-valued class $\mathcal{H}$, and every L-Lipschitz $h : (\mathbb{R}^d, \|\cdot\|) \to \mathbb{R}$, is $C = \kappa_{\|\cdot\|}^{-1}$.

Proof. The upper bound is Theorem 4.1. For the lower bound, take $n = 1$ and a class with two functions whose values at the single sample point are 0 and $v \in \mathbb{R}^d$. Let $h(u) = \|u\|$, which is 1-Lipschitz. Then

$$\widehat{\mathfrak{R}}_1(h \circ \mathcal{H}) = \mathop{\mathbb{E}}_{\varepsilon} \max\{0, \varepsilon \|v\|\} = \frac{\|v\|}{2},$$

while, by symmetry of $\langle \sigma, v\rangle$,

$$\widehat{\mathfrak{R}}_1(\mathcal{H}) = \mathop{\mathbb{E}}_{\sigma} \max\{0, \langle \sigma, v\rangle\} = \frac{1}{2} \mathop{\mathbb{E}}_{\sigma} |\langle \sigma, v\rangle| .$$

Thus any constant C satisfying (16) must obey

$$C \geq \frac{\|v\|}{\mathbb{E}_\sigma |\langle \sigma, v\rangle|}.$$

Taking the supremum over $v \neq 0$ gives $C \geq \kappa_{\|\cdot\|}^{-1}$. □

5 Explicit constants for ℓ_q^d

The exact value of $\kappa_{\|\cdot\|}$ may be inconvenient to compute. A simple explicit bound follows from the sharp $p = 1$ Khintchine inequality

$$\mathop{\mathbb{E}}_{\tau} |\langle \tau, v\rangle| \geq \frac{1}{\sqrt{2}} \|v\|_2 , \qquad v \in \mathbb{R}^d, \tag{17}$$

see Haagerup [3]. Let

$$\Delta_2(\|\cdot\|) := \sup_{\|v\|_2=1} \|v\| . \tag{18}$$

Then $\|v\| \leq \Delta_2(\|\cdot\|) \|v\|_2$, so

$$\kappa_{\|\cdot\|} \geq \frac{1}{\sqrt{2}\, \Delta_2(\|\cdot\|)}. \tag{19}$$

Consequently,

$$\widehat{\mathfrak{R}}_n(h \circ \mathcal{H}; x_{1:n}) \leq \sqrt{2}\, L\, \Delta_2(\|\cdot\|)\, \widehat{\mathfrak{R}}_n(\mathcal{H}; x_{1:n}). \tag{20}$$

For $\|\cdot\| = \|\cdot\|_q$ on $\mathbb{R}^d$,

$$\Delta_2(\|\cdot\|_q) = d^{(1/q-1/2)_+}, \qquad 1 \leq q \leq \infty, \tag{21}$$

where $1/\infty := 0$ and $a_+ := \max\{a, 0\}$. This gives

$$\widehat{\mathfrak{R}}_n(h \circ \mathcal{H}; x_{1:n}) \leq \sqrt{2}\, L\, d^{(1/q-1/2)_+} \widehat{\mathfrak{R}}_n(\mathcal{H}; x_{1:n}), \qquad 1 \leq q \leq \infty. \tag{22}$$

For $q = \infty$ the exact constant is better than (22).

Proposition 5.1 (The ℓ_∞ constant)**.** *For the full multivariate Rademacher complexity,*

$$\kappa_{\|\cdot\|_\infty} = 1. \tag{23}$$

Hence every L-Lipschitz map $h : (\mathbb{R}^d, \|\cdot\|_\infty) \to \mathbb{R}$ satisfies

$$\widehat{\mathfrak{R}}_n(h \circ \mathcal{H}; x_{1:n}) \leq L \widehat{\mathfrak{R}}_n(\mathcal{H}; x_{1:n}). \tag{24}$$

Proof. If $\|v\|_\infty = 1$, choose j with $|v_j| = 1$ and condition on all signs except τ_j. Writing $S = \sum_{k\neq j} \tau_k v_k$, we have

$$\mathop{\mathbb{E}}_{\tau_j} |S + \tau_j v_j| = \frac{|S + v_j| + |S - v_j|}{2} \geq |v_j| = 1.$$

Thus $\mathbb{E}_\tau |\langle \tau, v\rangle| \geq \|v\|_\infty$ for all v, so $\kappa_{\|\cdot\|_\infty} \geq 1$. The reverse inequality follows from $v = e_j$, for which $\mathbb{E} |\langle \tau, e_j\rangle| = 1 = \|e_j\|_\infty$. □

For $1 \leq q < 2$, a dimension factor is unavoidable for the standard coordinate-Rademacher comparison.

Proposition 5.2 (Unavoidable dimension dependence for $1 \leq q < 2$)**.** *For $1 \leq q < 2$,*

$$\kappa_{\|\cdot\|_q}^{-1} \geq d^{1/q-1/2}. \tag{25}$$

Therefore no bound of the form $\widehat{\mathfrak{R}}_n(h\circ\mathcal{H}) \leq C_q L \widehat{\mathfrak{R}}_n(\mathcal{H})$ with C_q independent of d can hold uniformly over all dimensions when the right-hand side is the standard coordinate-Rademacher complexity.

Proof. Take $v = (1, \ldots, 1)$. Then $\|v\|_q = d^{1/q}$ and, by Cauchy's inequality,

$$\mathop{\mathbb{E}}_{\tau} |\langle \tau, v\rangle| \leq \left(\mathop{\mathbb{E}}_{\tau} |\langle \tau, v\rangle|^2\right)^{1/2} = \sqrt{d}.$$

Thus $\kappa_{\|\cdot\|_q} \leq d^{1/2-1/q}$, which is (25). Proposition 4.2 converts this into the claimed lower bound for any universal contraction constant. □

6 Norm-adapted linear complexities

Theorem 3.1 is not limited to coordinate Rademacher vectors. It is a first-moment domination principle. If one is willing to replace the right-hand side by a linear complexity adapted to the norm, the same theorem can remove some dimension factors.

6.1 Stable comparison for $1 < q < 2$

Let $1 < q < 2$. Let Z be a nontrivial symmetric q-stable random variable, normalized in any fixed way, and put $m_q := \mathbb{E}\,|Z| < \infty$. Let $Y = (Z_1, \ldots, Z_d)$ have independent copies of Z as coordinates. By q-stability,

$$\langle Y, v\rangle = \sum_{j=1}^{d} Z_j v_j \overset{d}{=} \|v\|_q\, Z, \qquad \mathbb{E}\,|\langle Y, v\rangle| = m_q \|v\|_q\,. \tag{26}$$

Thus $\rho(Y; \|\cdot\|_q) = m_q$. Define

$$\widehat{\mathfrak{S}}_{q,n}(\mathcal{H}; x_{1:n}) := \mathop{\mathbb{E}}_{Z}\left[\sup_{f\in\mathcal{H}} \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{d} Z_{ij} f_j(x_i)\right], \tag{27}$$

whenever the expectation is finite; otherwise the inequality below is read in the extended-real sense. Corollary 3.2 gives

$$\widehat{\mathfrak{R}}_n(h\circ\mathcal{H}; x_{1:n}) \le \frac{L}{m_q}\widehat{\mathfrak{S}}_{q,n}(\mathcal{H}; x_{1:n}). \tag{28}$$

If Z is scaled so that $m_q = 1$, the contraction constant is L. This is in the same spirit as the p-stable extension of Maurer's theorem due to Zatarain-Vera [5]. The point here is that it is an immediate consequence of the general first-moment domination theorem.

6.2 Coordinate sampling for ℓ_1

Although a 1-stable variable has infinite first absolute moment, finite ℓ_1^d has an exact elementary representation. Let J be uniform on $\{1, \ldots, d\}$, let η be a scalar Rademacher sign, and put $Y = d\eta e_J$. Then

$$\mathop{\mathbb{E}}_{J,\eta}|\langle Y, v\rangle| = \frac{1}{d}\sum_{j=1}^{d} d\,|v_j| = \|v\|_1\,. \tag{29}$$

Consequently, every L-Lipschitz $h : (\mathbb{R}^d, \|\cdot\|_1) \to \mathbb{R}$ satisfies

$$\widehat{\mathfrak{R}}_n(h\circ\mathcal{H}; x_{1:n}) \le L \mathop{\mathbb{E}}_{J,\eta}\left[\sup_{f\in\mathcal{H}} \frac{1}{n}\sum_{i=1}^{n} d\eta_i f_{J_i}(x_i)\right]. \tag{30}$$

This is a different right-hand side from the standard coordinate-Rademacher complexity. It is often the natural price of obtaining a dimension-free constant in an ℓ_1 Lipschitz problem.

7 Implication for predict-then-optimize bounds

We now record the precise way in which the theorem plugs into a bounded-loss generalization argument. Let $Z = (X, C)$ denote an input-cost pair and suppose $Z_1, \ldots, Z_n$ are i.i.d. For $f \in \mathcal{H}$, write

$$g_f(x,c) := \ell(f(x),c), \qquad R_\ell(f) := \mathbb{E}\, g_f(Z), \qquad \widehat{R}_\ell(f) := \frac{1}{n}\sum_{i=1}^n g_f(Z_i).$$

Assume $0 \le g_f \le B$ for all f, and assume that for each realized cost c, the map $u \mapsto \ell(u,c)$ is L-Lipschitz with respect to a norm $\|\cdot\|_{\mathrm{cost}}$ on the prediction/cost-vector space. In the notation of El Balghiti et al., this is typically the dual norm on costs when the feasible-set geometry is expressed in a primal decision norm.

Let

$$\mathfrak{R}_n(\ell \circ \mathcal{H}) := \mathop{\mathbb{E}}_{Z_{1:n}} \mathop{\mathbb{E}}_{\varepsilon} \sup_{f\in\mathcal{H}} \frac{1}{n}\sum_{i=1}^n \varepsilon_i \ell(f(X_i), C_i)$$

be the expected scalar Rademacher complexity, and let

$$\mathfrak{R}_n(\mathcal{H}) := \mathop{\mathbb{E}}_{X_{1:n}} \mathop{\mathbb{E}}_{\sigma} \sup_{f\in\mathcal{H}} \frac{1}{n}\sum_{i=1}^n \langle \sigma_i, f(X_i)\rangle$$

be the expected standard multivariate Rademacher complexity of the prediction class. The sample-wise contraction theorem implies

$$\mathfrak{R}_n(\ell \circ \mathcal{H}) \le \frac{L}{\kappa_{\|\cdot\|_{\mathrm{cost}}}} \mathfrak{R}_n(\mathcal{H}). \tag{31}$$

Combining (31) with the standard bounded-loss Rademacher inequality of Bartlett and Mendelson [1] gives the following.

Corollary 7.1 (Bounded-loss plug-in bound)**.** *Under the assumptions in the preceding paragraph, with probability at least $1-\delta$ over the sample $Z_1,\dots,Z_n$, every $f \in \mathcal{H}$ satisfies*

$$R_\ell(f) \le \widehat{R}_\ell(f) + \frac{2L}{\kappa_{\|\cdot\|_{\mathrm{cost}}}}\mathfrak{R}_n(\mathcal{H}) + B\sqrt{\frac{\log(1/\delta)}{2n}}. \tag{32}$$

Remark 7.2 (Why this is deliberately stated with expected complexity)**.** *The high-probability statement (32) uses expected Rademacher complexity. One can state empirical, data-dependent variants, but they require the usual additional concentration terms or adjusted constants. Thus it would be misleading to simply replace $\mathfrak{R}_n(\mathcal{H})$ in (32) by the realized empirical quantity $\widehat{\mathfrak{R}}_n(\mathcal{H}; X_{1:n})$ without further work.*

This is the promised partial progress for the predict-then-optimize setting: if the margin loss used in the paper is Lipschitz in a non-Euclidean cost norm, then the Euclidean contraction step can be replaced by (31) and the resulting generalization bound has the norm-dependent factor $\kappa^{-1}_{\|\cdot\|_{\mathrm{cost}}}$, or the corresponding norm-adapted Y-linear complexity if one chooses a different dual random vector Y.

8 Conclusion

The main mathematical output is the first-moment domination principle, Theorem 3.1. It completely solves the standard finite-dimensional arbitrary-norm vector-contraction subproblem:

$$\widehat{\mathfrak{R}}_n(h \circ \mathcal{H}) \le C(\|\cdot\|) L \widehat{\mathfrak{R}}_n(\mathcal{H}), \qquad C(\|\cdot\|) = \left(\min_{\|v\|=1} 2^{-d} \sum_{\tau\in\{\pm 1\}^d} |\langle \tau, v\rangle|\right)^{-1}.$$

This constant is finite for every norm on $\mathbb{R}^d$ and is sharp for the standard coordinate-Rademacher comparison. For ℓ_q^d, the explicit upper bound is $C(\|\cdot\|_q) \leq \sqrt{2}\, d^{(1/q-1/2)_+}$, with the exact improvement $C(\|\cdot\|_\infty) = 1$. For $1 \leq q < 2$, dimension-free constants are impossible for the standard comparison; dimension-free statements require a norm-adapted linear complexity, such as a stable complexity for $1 < q < 2$ or the coordinate-sampling complexity for $q = 1$.

Thus the result is a solution of the contraction subproblem and partial progress for the authors' predict-then-optimize paper.

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